

Sourodeep Roy

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Profile

MSc Data Science & Analytics graduate (Distinction, University of Leeds) with two years of production data engineering and analytics experience at Cognizant. Background spans statistical modelling, machine learning, and applied NLP, with project experience in end-to-end ML system development, LLM pipeline design, and quantitative risk modelling. Comfortable working with large datasets in Python and SQL, and focused on building production-grade machine learning and artificial intelligence systems.

Education

University of Leeds MSc Data Science & Analytics — Distinction <i>Relevant: Statistical Learning, Time Series Analysis, Machine Learning, Deep Learning, Optimisation</i>	<i>Sept 2024 – Sept 2025</i> <i>Leeds, UK</i>
University of Engineering and Management, Kolkata BTech Computer Science & Engineering — CGPA: 8.56/10	<i>Jul 2018 – Jun 2022</i> <i>Kolkata, India</i>

Quantitative Projects

Credit Risk Intelligence Suite

Python | LLM | FinBERT | SHAP | XGBoost | WoE/IV | FRED API | Ollama

Extended a production credit default model on 1.3M Lending Club loans with three GenAI layers — LLM signal extraction, SHAP-grounded credit memo generation, and a macro-driven Early Warning System — evaluating each component rigorously against held-out loan performance data.

- Trained a WoE logistic regression scorecard on 14 features with temporal validation (2007–2015 train, 2016–2018 test); AUROC 0.767, Gini 0.534; XGBoost challenger gained only +0.010 Gini, confirming logistic regression as champion on interpretability and regulatory grounds.
- Deployed a local LLM pipeline (Ollama/llama3.2:3b) to extract five structured risk signals from 120,000 borrower descriptions at 99.9% parse rate; signals used to enrich individual loan risk profiles with borrower intent, financial distress indicators, and repayment confidence alongside structured WoE features.
- Built a SHAP-grounded credit memo generator producing audit-ready four-section analyst memos; hallucination evaluation across 200 loans confirmed SHAP consistency 100%, non-invention rate 98%; Virtual Expert query layer achieved 94% grounding rate and 100% factual accuracy across 50 automated analyst queries.
- Constructed a two-stream Early Warning System combining FinBERT news sentiment with FRED macro signals (unemployment, credit card delinquency, fed funds rate, consumer sentiment); backtested against 2016–2018 sector default rates achieving recall 1.00 across five sectors, with the macro stream lifting recall over news-only by +0.18.

Intraday Volatility & Market Risk Forecasting using ACD-GARCH

MSc Dissertation — Distinction (80%) | R | Weibull ACD(2,2) | GARCH Modelling | VaR/ES Backtesting

Investigated whether explicitly modelling trade arrival intensity improves intraday volatility forecasting and tail-risk calibration over standard GARCH applied to raw returns. Built on BP plc LSE tick quotes (Apr–Jun 2020); all models trained on April only and evaluated on May–June without refitting for out-of-sample validation.

- Engineered a microstructure-consistent event series from millisecond quote data — filtering to LSE trading hours, defining events as midquote changes, and removing intraday seasonality from durations using an April-calibrated median diurnal profile.
- Estimated a Weibull ACD(2,2) on event durations to recover conditional intensity; used the intensity process to construct adjusted returns that isolate genuine price volatility from activity-driven noise before GARCH estimation.
- Benchmarked sGARCH, EGARCH, and GJR-GARCH with Gaussian and Student-t innovations across both raw and intensity-adjusted series; intensity-adjusted models achieved 69–72% lower MSE and 4.8–5.7% better QLIKE out-of-sample, with Diebold-Mariano tests confirming statistical significance and VaR/ES backtesting confirming superior tail calibration.

Professional Experience

Cognizant <i>Programmer Analyst — AI & Analytics</i>	<i>Aug 2022 – Jul 2024</i>
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- Built production-grade data and analytics pipelines in Python and PySpark/Spark SQL across 20+ heterogeneous sources, translating business requirements into robust, testable workflows used for downstream reporting and decisioning; improved ingestion latency by 40% and query performance by 25%.
- Implemented 100+ automated data-quality and integrity checks (schema validation, reconciliation, anomaly detection, failure alerting) with end-to-end lineage traceability; reduced downstream discrepancies by 30% — directly applicable to model input integrity and data remediation in credit risk workflows.
- Orchestrated end-to-end workflows using Apache Airflow with logging, observability, and incident handling; automated 80% of routine processing and sustained 97% pipeline uptime. Designed modular, reusable Python components for ingestion, transformation, and validation aligned with maintainability and audit standards.

Technical Skills

Programming	Python, R, SQL, C++
ML & Statistics	Linear/Logistic/Ridge/Lasso Regression, Decision Trees, Random Forests, XGBoost, LightGBM, SVMs, K-NN, SHAP, WoE/IV Feature Engineering, Time-Series (ARIMA, GARCH Family, ACD Models), MLE, Hypothesis Testing
Deep Learning	LSTMs, CNNs, Transformers, PyTorch; Sequence Modelling, Classification, Feature Extraction
GenAI & NLP	LLM Prompt Engineering, Local Inference (Ollama), FinBERT, Sentence Transformers, TF-IDF, VADER, LangChain, Groq API, Hallucination Evaluation
Libraries	Pandas, NumPy, SciPy, scikit-learn, statsmodels, Transformers (HuggingFace), SHAP, Matplotlib, Seaborn, rugarch, xts, zoo